TITANIC SURVIVAL PREDICTION

MACHINE LEARNING PROJECT REPORT

Submitted by

SANJANA JAIN

Project Title: Titanic Survival Prediction

Technology Used: Python, Google Colab, Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn

Machine Learning Algorithm: Random Forest Classifier

Project Type: Internship Project

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported and guided me in completing my project, **“Titanic Survival Prediction.”**

This project provided me with an opportunity to learn and apply machine learning concepts using Python and Google Colab. I gained practical knowledge of data preprocessing, exploratory data analysis, model training, and model evaluation.

I am thankful to my mentors and everyone who helped me during the development of this project.

Finally, I would like to thank all those who contributed directly or indirectly to the successful completion of this project.

Sanjana Jain

ABSTRACT

The Titanic Survival Prediction project is a machine learning project developed using Python in Google Colab. The main purpose of this project is to predict whether a passenger survived the Titanic disaster based on passenger information.

The project uses the Titanic dataset obtained from Kaggle. The dataset is explored and processed to prepare it for machine learning. Exploratory Data Analysis (EDA) is performed to understand the data and identify patterns. A Random Forest Classifier is then trained to predict passenger survival.

The model is evaluated using a training and validation split. It achieved 82.12% validation accuracy on the selected validation data. This result indicates that the model was able to classify passenger survival with reasonable accuracy on that split.

This project provides practical experience in data preprocessing, data visualization, classification, and model evaluation. It also demonstrates how machine learning can be applied to a real-world historical dataset.

1. INTRODUCTION

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions. It is widely used in different fields, such as healthcare, banking, education, transportation, and data analysis.

The Titanic Survival Prediction project focuses on predicting whether a passenger survived the Titanic disaster of 1912. The prediction is based on passenger details available in the dataset, such as age, sex, passenger class, fare, and other relevant information.

In this project, Python is used for data processing, analysis, visualization, and model development. Google Colab provides an online environment for writing and executing Python code. Libraries such as Pandas and NumPy help process the data, while Matplotlib and Seaborn are used to visualize patterns. Scikit-learn is used to train and evaluate the machine learning model.

A Random Forest Classifier is used to predict passenger survival. The model is evaluated using a validation dataset, and it achieves 82.12% validation accuracy on the selected split.

This project demonstrates the practical application of machine learning in a classification problem and helps develop an understanding of the complete machine learning workflow, from data exploration to model evaluation.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

The Titanic disaster resulted in the loss of many lives. Passenger survival depended on several factors, including passenger class, age, sex, and other characteristics.

The problem addressed in this project is to use machine learning techniques to predict whether a passenger survived or did not survive based on the available passenger information. This is a binary classification problem because the model predicts one of two possible outcomes.

2.2 Objectives

The main objectives of this project are:

1. To understand and explore the Titanic dataset.
2. To identify important features related to passenger survival.
3. To perform data cleaning and handle missing values.
4. To visualize the data using graphs and charts.
5. To prepare the dataset for machine learning.
6. To train a Random Forest Classifier.
7. To evaluate the model using validation accuracy.
8. To generate predictions for passenger survival.
9. To gain practical experience in Python and machine learning.

2.3 Scope of the Project

This project focuses on building and evaluating a machine learning classification model using the Titanic dataset. It demonstrates the steps involved in a typical machine learning workflow, including data exploration, preprocessing, model training, and evaluation. The model's predictions are based on the information available in the dataset and are not guaranteed to be correct for every passenger.

3. DATASET DESCRIPTION

3.1 Dataset Overview

The Titanic dataset used in this project is obtained from Kaggle. It contains information about passengers who were aboard the RMS Titanic. The dataset includes passenger details and a survival label that can be used to train a machine learning classification model.

The training dataset is used to explore passenger information and train the model. The test dataset is used to generate survival predictions for passengers whose survival labels are not provided.

3.2 Important Dataset Features

The dataset contains several columns that describe passengers and their travel details. Important features include:

1. **PassengerId:** A unique identification number assigned to each passenger.
2. **Survived:** The target variable indicating survival (0 = did not survive, 1 = survived).
3. **Pclass:** The passenger's ticket class (1st, 2nd, or 3rd class).
4. **Name:** The passenger's name.
5. **Sex:** The passenger's recorded sex.
6. **Age:** The passenger's age.
7. **SibSp:** The number of siblings or spouses aboard.
8. **Parch:** The number of parents or children aboard.
9. **Ticket:** The passenger's ticket number.
10. **Fare:** The passenger's ticket fare.
11. **Cabin:** The cabin information, where available.
12. **Embarked:** The port where the passenger boarded.

3.3 Target Variable

The target variable in the training dataset is **Survived**. It has two possible values: 0 and 1. A value of 0 represents a passenger who did not survive, while a value of 1 represents a passenger who survived.

The machine learning model learns patterns from the available features to predict this target variable.

3.4 Dataset Source

The dataset is available through the Kaggle Titanic competition:

https://www.kaggle.com/competitions/titanic/data

4. TOOLS AND TECHNOLOGIES USED

The Titanic Survival Prediction project was developed using Python and various data analysis and machine learning libraries. These tools helped in data processing, visualization, model training, and evaluation.

4.1 Python

Python is a high-level programming language widely used for data analysis and machine learning. It was used to write the code and implement the project workflow.

4.2 Google Colab

Google Colab is an online platform that allows users to write and execute Python code in notebook format. It was used as the development environment for this project.

4.3 Pandas

Pandas is a Python library used for loading, organizing, and manipulating structured data. It helped in reading the Titanic dataset and examining its columns.

4.4 NumPy

NumPy is a Python library that supports numerical operations and efficient handling of arrays and numerical data.

4.5 Matplotlib and Seaborn

Matplotlib and Seaborn are visualization libraries used to create charts and graphs. They helped explore patterns and relationships in the Titanic dataset.

4.6 Scikit-learn

Scikit-learn is a machine learning library that provides tools for data preprocessing, model training, and evaluation. It was used to build and assess the Random Forest classification model.

4.7 Random Forest Classifier

Random Forest is a supervised machine learning algorithm that combines multiple decision trees to make predictions. It was selected for this project to classify passengers according to their predicted survival outcome.

5. PROJECT METHODOLOGY

The project follows a systematic machine learning workflow, starting from data loading and exploration and continuing through preprocessing, model training, and evaluation.

5.1 Data Collection

The Titanic dataset was obtained from Kaggle and loaded into Google Colab using Python. The dataset was used to explore passenger information and build a survival prediction model.

5.2 Data Exploration

The dataset was examined to understand its structure, column names, data types, and missing values. Descriptive statistics and visualizations were used to explore passenger characteristics and survival patterns.

5.3 Data Preprocessing

The data was prepared for machine learning by handling missing values and selecting relevant input features. Categorical information was converted into a suitable numerical representation where required.

5.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed using graphs and charts. These visualizations helped examine relationships between survival and passenger characteristics such as sex, age, and passenger class.

5.5 Splitting the Dataset

The available labeled training data was divided into training and validation subsets. The training subset was used to fit the model, while the validation subset was used to evaluate its predictive performance.

5.6 Model Training

A Random Forest Classifier was trained using the prepared training data. The algorithm combines predictions from multiple decision trees to produce a classification result.

The trained model was evaluated on the validation subset using accuracy. The model achieved 82.12% validation accuracy on the selected split.

5.8 Prediction Generation

After model development, the model was used to generate survival predictions for the separate Titanic test dataset, where applicable. These predictions indicate whether each **5.7 Model Evaluation**

passenger is predicted to have survived or not survived.

6. SYSTEM IMPLEMENTATION

The Titanic Survival Prediction project was implemented using Python in Google Colab. The implementation involved loading the dataset, preparing the data, training a machine learning model, and evaluating its performance.

6.1 Importing Libraries

Python libraries were imported to perform data manipulation, visualization, preprocessing, and machine learning tasks. Pandas and NumPy were used for data operations, while Matplotlib and Seaborn supported data visualization. Scikit-learn was used for model development and evaluation.

6.2 Data Preparation

The dataset was loaded and examined to understand its structure. The input features and target variable were prepared for model training. Preprocessing steps were applied as required to handle missing values and categorical data.

6.3 Model Development

A Random Forest Classifier was used to build the survival prediction model. This algorithm combines multiple decision trees to generate a classification prediction.

6.4 Model Training and Validation

The model was trained using the training data. A separate validation subset was used to evaluate how well the model predicted passenger survival.

6.5 Performance Evaluation

The model achieved **81.56% validation accuracy** on the selected validation split. This result provides an estimate of model performance on the validation data used in this project.

7. RESULTS AND EVALUATION

The Titanic Survival Prediction model was evaluated using a separate validation subset of the available training data. The purpose of the evaluation was to measure how accurately the model predicted passenger survival.

7.1 Model Accuracy

The Random Forest Classifier achieved **81.56% validation accuracy** on the selected validation split.

Accuracy measures the proportion of predictions that were correct out of all predictions made on the validation data.

```
Validation accuracy: 81.56%

Classification Report:
              precision    recall  f1-score   support

           0       0.82      0.89      0.86       110
           1       0.80      0.70      0.74        69

    accuracy                           0.82       179
   macro avg       0.81      0.79      0.80       179
weighted avg       0.81      0.82      0.81       179
```

7.2 Result Analysis

The validation result indicates that the model learned useful patterns from the training data and was able to predict survival outcomes for the validation samples.

However, accuracy alone does not provide a complete picture of model performance. Additional evaluation measures, such as precision, recall, F1-score, and a confusion matrix, can provide further insight into the model's predictions.

7.3 Overall Outcome

The project successfully demonstrates the application of a Random Forest classification model to the Titanic survival prediction problem. It also provides practical experience in data preprocessing, exploratory data analysis, model training, and validation.

8. CONCLUSION

The Titanic Survival Prediction project was successfully completed using Machine Learning techniques in Google Colab. The main objective of this project was to predict whether a passenger survived the Titanic disaster based on passenger information.

The project involved data collection, data cleaning, exploratory data analysis, data preprocessing, model training, and model evaluation. The Random Forest Classifier was used to train the machine learning model.

The model achieved a validation accuracy of **81.56%**. This result shows that the model was able to predict passenger survival with reasonable accuracy on the validation data.

This project helped in understanding the practical implementation of machine learning and the importance of data preprocessing and model evaluation. In the future, the project can be improved by testing other algorithms, tuning model parameters, and using additional evaluation techniques.

9. FUTURE SCOPE

The Titanic Survival Prediction project can be improved in several ways in the future.

1. **Testing More Algorithms:** Other machine learning algorithms, such as Logistic Regression, Decision Tree, and Support Vector Machine, can be tested and compared with the Random Forest Classifier.
2. **Improving Model Accuracy:** Hyperparameter tuning and better feature selection can be used to improve the model's performance.
3. **Data Visualization:** More graphs and charts can be added to understand passenger survival patterns more clearly.
4. **User-Friendly Interface:** A web application can be developed where users enter passenger details and receive a survival prediction.
5. **Model Evaluation:** Additional evaluation measures, such as precision, recall, F1-score, and a confusion matrix, can be used to understand the model's performance in greater detail.

This project provides a foundation for learning and applying machine learning techniques to classification problems.

10. REFERENCES

The following resources were used for developing this project:

1. **Kaggle Titanic Dataset**
   https://www.kaggle.com/competitions/titanic/data
2. **Python Documentation**
   https://docs.python.org/3/
3. **Pandas Documentation**
   https://pandas.pydata.org/docs/
4. **NumPy Documentation**
   https://numpy.org/doc/
5. **Scikit-learn Documentation**
   https://scikit-learn.org/stable/
6. **Google Colab**
   https://colab.research.google.com/

These resources helped in understanding the dataset, performing data preprocessing, building the machine learning model, and evaluating its performance.